

UCB–Pandora Stopping for Generation with Unknown Value Distributions and Unknown Response Costs

Abstract

We study adaptive generation when both the prompt-specific value distribution and the character cost of the next response are unknown. A policy opens a small pilot prefix, fits a training-regularized upper-tail model, estimates the next response length from information available at that prefix, and continues only while an adjusted estimate of one-step improvement exceeds the estimated cost. Positive-width configurations use a UCB-style optimistic deformation; some coding target/equal-budget configurations use the zero-width plug-in boundary. We give one shared UCB–Pandora blueprint and trace two concrete implementations—alignment with a Bradley–Terry link and coding with train-only isotonic probability calibration—to that blueprint. We also separate the economic utility divisor from a selected reservation divisor in the improved coding utility experiment. The description below follows the retained source code and saved experiment artifacts, including their tuning protocols, finite horizons, calibration protocol, evaluation boundaries, and evaluation-only oracle comparisons.

1 Unknown-Distribution, Unknown-Cost Pandora Stopping

1.1 Problem definition

Fix a prompt or programming problem x . Its finite candidate pool is exposed in a random order, so the generations act as exchangeable “boxes.” Opening the j th box reveals both a raw reward-model score R_j and an output length C_j in characters. Neither realization is known before the box is opened, and the policy is not given the prompt-specific law of either quantity. This is the precise sense in which both the value distribution and the opening cost are unknown.

Let $\mathcal{A}$ denote the training and development data available before final-test evaluation, and let $\mathcal{F}_n$ denote the information generated by $\mathcal{A}$ and the first n observed score–length pairs. A decision at prefix n may depend only on $\mathcal{F}_n$. Define

$$V_n = \max_{j \leq n} u_n(R_j), \quad J_n = \min \arg \max_{j \leq n} u_n(R_j), \quad (1)$$

where u_n maps a score to the utility scale used by the stopping rule. The map is prefix-dependent in alignment and frozen in coding. Ties in coding are resolved by the first maximum, exactly as in the implementation. Let $\mathcal{U}_x(j)$ be the utility used in the primary utility evaluation: full-pool benchmarked BT quality for alignment and binary correctness for coding. For a supplied utility divisor D_{util} , its primary objective is

$$\mathbb{E} \left[\mathcal{U}_x(J_T) - \lambda_{\text{util}} \sum_{j=1}^T C_j \right], \quad \lambda_{\text{util}} = \frac{1}{D_{\text{util}}}. \quad (2)$$

Here T is an $\mathcal{F}_n$ -stopping time. Thus the price λ_{util} of one character is known and externally specified. The unknown opening cost is the next response’s length C_{n+1} , or, equivalently, its utility cost C_{n+1}/D_{util} . Larger D_{util} makes characters cheaper and ordinarily induces more openings. The first $n_0 = 3$ boxes are always opened, so their realized cost is unavoidable under the implemented policy class. Later experiments also report quality at matched cost and cost at matched quality; Section 1.6 states explicitly when those comparisons use held-out outcomes only for descriptive matching.

For comparison, under the classical assumptions of independent boxes, recall, risk-neutral utility, a known homogeneous value law F , and known deterministic opening cost c , the reservation test continues from incumbent v when

$$\text{EI}_F(v) = \mathbb{E}_{W \sim F}[(W - v)_+] > c. \quad (3)$$

Equivalently, the reservation value z solves $\mathbb{E}[(W - z)_+] = c$. Our policies replace both unknown sides of this test: they estimate a training-regularized, optionally optimistic value law from pre-test data and the opened prefix, and estimate the mean next length without looking at the unopened suffix. The resulting rules are empirical reservation policies. They do not claim the exact optimality of classical Pandora’s rule: the finite pool is sampled without replacement, the fitted tail can be misspecified, and the code models reward and length marginals separately.

1.2 Calibration and utility–cost scales

Before specifying either tail model, we explain how reward and character count are placed on a common scale. Raw reward and response length are not commensurate. Each implementation first maps reward to a quantity in $[0, 1]$, then divides predicted characters by a reservation divisor D_{res} . Its stopping test therefore compares expected improvement with estimated opening cost on a common reservation scale. In the original experiments $D_{\text{res}} = D_{\text{util}}$; the improved coding utility experiment deliberately selects D_{res} separately from the fixed economic price D_{util} in Eq. (2). Target-quality experiments have no economic utility price: there D_{res} is simply a stopping parameter chosen to meet a quality constraint cheaply. The two reward-to-value maps also have different statistical meanings.

Alignment: a Bradley–Terry-style link. At prefix n , alignment uses $u_n(r) = \sigma(r - b_n)$, where b_n is an online estimate of a high-quality reward benchmark. Under a unit-temperature Bradley–Terry model, this is the probability assigned to score r beating a benchmark with score b_n . It is a *relative* win probability with a moving reference, not an absolute probability learned from binary labels. The implementation fits neither a BT temperature nor pairwise calibration parameters; “BT” names the fixed sigmoid link. Because that link is strictly increasing, the induced action is still to return the opened generation with largest raw reward.

Coding: train-only probability calibration. Coding instead fits one nondecreasing isotonic map $\hat{g}_s(r) \approx \Pr(\text{correct} \mid R = r)$ on all labeled generations from the outer-training problems of split s . This is an *absolute* success-probability estimate pooled over those training problems. After fitting, the map is frozen and applied pointwise to training and outer-test scores. The split- s isotonic fitting call consumes only its 41 outer-training problems; the target selector’s additional development-profile provenance is discussed with that experiment below. The map is not a task-conditional oracle and need not remain perfectly calibrated after a distribution shift. The shifted-exponential law is fit directly in this bounded probability space; the retained coding results do not apply a BT transform after isotonic calibration. Correctness labels fit this map and score final predictions, but the online policy never observes them. Generating a candidate reveals only the program, its reward-model score, and its length.

Cost conversion. Both methods compare their expected utility gain with $\hat{\mu}_n^C / D_{\text{res}}$, where $\hat{\mu}_n^C$ estimates the next response length. This estimate controls stopping only. Every reported experiment charges the exact prefix sum $\sum_{j \leq \tau} C_j$, never the estimate. Neither method puts a confidence bound on cost: “UCB” refers to optimism on the value side, with a plug-in estimate on the cost side.

The phrase *held-out calibration* therefore requires care. The coding isotonic map is held out from the final outer test, but it is fit once on all 41 outer-training problems and is not refit inside each fold of the four-fold policy-selection calculation. Alignment has no labeled held-out calibration map: its BT benchmark is recomputed from the observed prefix. A full-pool benchmark appears only in alignment evaluation and is never passed to the stopping rule.

1.3 A shared high-level algorithm

At every observed prefix, both methods perform the same three operations: they estimate the utility upside of one more response from an upper-tail model, estimate that response’s length, and open it only when adjusted expected improvement exceeds predicted character cost. The task-specific choices enter through the utility map, tail fit, and gain adjustment.

Both implementations compute expected improvement from a fitted conditional upper tail, weight that conditional expectation by the configured mass $p = 1 - q$, and assign zero improvement to the remaining mass in the implemented approximation. Let $\tilde{W}_n^+$ be a future decision-utility draw conditional on that fitted, optionally optimistic tail, and let $\hat{\mu}_n^C$ estimate the mean length of response $n + 1$ from $\mathcal{F}_n$. Their common expected-improvement estimate is

$$\widehat{\text{EI}}_n = p \mathbb{E}[(\tilde{W}_n^+ - V_n)_+ \mid \mathcal{F}_n], \quad G_n = \Psi_n(\widehat{\text{EI}}_n), \quad (4)$$

Algorithm 1 Unknown-distribution, unknown-cost Pandora blueprint

Require: Training data $\mathcal{A}$; reservation divisor D_{res} ; minimum openings n_0 ; horizon H

- 1: Fit a value map (if needed) and distribution/cost priors using only pre-test data $\mathcal{A}$
 - 2: Open n_0 candidates and observe their scores and exact lengths
 - 3: $n \leftarrow n_0$
 - 4: **while** $n < H$ **do**
 - 5: From $\mathcal{F}_n$, construct u_n , a fitted, optionally optimistic tail $(p, \widetilde{W}_n^+)$, and $\widehat{\mu}_n^C$
 - 6: $V_n \leftarrow \max_{j \leq n} u_n(R_j)$
 - 7: $\widehat{\text{EI}}_n \leftarrow p\mathbb{E}[(\widetilde{W}_n^+ - V_n)_+ \mid \mathcal{F}_n]$
 - 8: $G_n \leftarrow \Psi_n(\widehat{\text{EI}}_n)$
 - 9: **if** $G_n \leq \widehat{\mu}_n^C/D_{\text{res}}$ **then**
 - 10: **break**
 - 11: **end if**
 - 12: Open candidate $n + 1$; observe (R_{n+1}, C_{n+1}) ; $n \leftarrow n + 1$
 - 13: **end while**
 - 14: **return** the incumbent opened candidate and charge $\sum_{j=1}^n C_j$ characters
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where Ψ_n is the implementation-specific gain adjustment. Alignment uses

$$\Psi_n^{\text{align}}(z) = \min \left\{ z + \eta \sqrt{\frac{\log(1/\delta)}{2n}}, 1 \right\}, \quad \Psi_n^{\text{code}}(z) = \left(\frac{n}{3}\right)^{-\gamma} z. \quad (5)$$

The policy continues exactly when $G_n > \widehat{\mu}_n^C/D_{\text{res}}$. Both methods instantiate Algorithm 1.

No independence between next value and next length is required for the risk-neutral one-step identity

$$\mathbb{E}[(W - V_n)_+ - C/D_{\text{res}} \mid \mathcal{F}_n] = \mathbb{E}[(W - V_n)_+ \mid \mathcal{F}_n] - \mathbb{E}[C \mid \mathcal{F}_n]/D_{\text{res}}. \quad (6)$$

The code nevertheless estimates the two marginal terms separately.

When $D_{\text{res}} \neq D_{\text{util}}$, the same decision can be written on the economic scale by comparing $(D_{\text{res}}/D_{\text{util}})G_n$ with $\widehat{\mu}_n^C/D_{\text{util}}$. This algebra is useful for the improved coding utility policy: changing the reservation divisor calibrates stopping without changing the objective used to score the stopped trajectory.

The expectation in Algorithm 1 is in decision-utility space; in the concrete equations below we write the raw or transformed future draw explicitly and apply u_n inside the integrand. The decision is non-anticipating: every entry used at prefix n depends only on the first n scores and lengths. Alignment shares each observable-prefix fit across the requested character divisors, but this produces exactly the same stops as independent sequential runs. If another box is opened, all prefix-dependent quantities are updated. The reward tail and mean next length are modeled separately; their joint dependence is neither estimated nor exploited. The decision utility need not equal the evaluation utility. In particular, alignment computes EI around a changing prefix benchmark, whereas its reported quality uses a full-pool benchmark available only after stopping. The former is therefore a deployable surrogate for the latter, not an illicit look-ahead.

Table 1: Road map from the shared reservation blueprint to the two retained instantiations. “Prefix” means recomputed from opened candidates only.

Component	Alignment	Coding
Decision utility	Prefix BT-style win proxy $\sigma(R - b_n)$	Frozen isotonic correctness probability $\hat{g}_s(R)$
Tail-model space	$\exp(R)$; unbounded shifted-exponential upper tail	$\hat{g}_s(R) \in [0, 1]$; shifted-exponential upper tail truncated at 1
Active training fit	Median tail scale/location ratio ρ	Isotonic map, tail location/scale priors, and mean characters
Gain adjustment	Upper tail scale, lower benchmark scale, and additive bonus	Optional upper location/scale, multiplicative tail decay, and utility-mode reservation calibration
Next-cost estimate	Prefix mean characters ($\kappa_c = 0$)	Prefix/prior shrinkage with $\kappa_c \in \{0, 10\}$
Hard horizon	960, the available pool	512 in primary utility/target modes; retained equal-budget mode may use a training-chosen cap
Online observation	Raw reward and response characters	Raw reward mapped through frozen $\hat{g}_s$, and response characters; not correctness
Evaluation outcome	Full-pool-benchmarked reward quality	Binary program correctness
Tie behavior	Raw-reward maximization	First opened calibrated-probability maximizer

We now instantiate each object in Algorithm 1, beginning with alignment.

1.4 Alignment instantiation: exponentiated-tail UCB–Pandora with BT-style utility

What is fit. Each generator contains 100 prompts with 960 generations per prompt. Within each of ten split IDs 30–39, 50 prompts are training prompts and 50 are test prompts. For training prompt i , transform every raw reward as $E_{ij} = \exp(\text{clip}(R_{ij}, -50, 50))$ and compute

$$a_i = Q_{0.5}(E_{i1}, \dots, E_{i,960}), \quad (7)$$

$$s_i = \max\{\text{mean}_{j:E_{ij} \geq a_i} E_{ij} - a_i, \epsilon\}, \quad (8)$$

$$\rho = \text{median}_{i \in \mathcal{A}} \frac{s_i}{\max(a_i, \epsilon)}. \quad (9)$$

Only the relative tail-scale prior ρ in Eq. (9) affects the retained policy. Other stored prior fields are inactive under the retained settings and are intentionally omitted. The retained runner holds the following configuration fixed across all evaluated splits and generators:

$$q = 0.5, \quad \kappa_r = 5, \quad \kappa_c = 0, \quad \alpha = 0.4, \quad \delta = 0.05, \quad \eta = 0.002, \quad n_0 = 3. \quad (10)$$

Here and below, $\epsilon = 10^{-8}$. The benchmark-calibration weight is also fixed at zero, so the policy uses the theoretical tail coefficient $-\log\{(1 - 0.99)/(1 - q)\} = \log 50$ rather than the stored empirical coefficient.

Online fit and utility. At prefix n , let T_n contain the largest $m_n = n - \lceil qn \rceil + 1$ values among $E_1, \dots, E_n$. The inclusive threshold convention is deliberate. It differs slightly from the interpolated NumPy quantile used when fitting the prior: at $q = 0.5$ and even n , the online threshold is the lower middle order statistic and T_n contains $n/2 + 1$ values. Define

$$\ell_n = \min T_n, \quad \tilde{s}_n = \max\{\text{mean}(T_n) - \ell_n, \epsilon\}, \quad (11)$$

$$s_n = \frac{m_n \tilde{s}_n + \kappa_r \ell_n \rho}{m_n + \kappa_r}, \quad h_n = \sqrt{\frac{\log(1/\delta)}{m_n}}, \quad (12)$$

$$s_n^+ = s_n(1 + \alpha h_n), \quad s_n^- = \max\{s_n(1 - \alpha h_n), \epsilon\}. \quad (13)$$

The fitted future upper tail has mass $1 - q = 0.5$ and $E = \ell_n + s_n^+ Z$ for $Z \sim \text{Exp}(1)$. The online proxy for the raw-reward 99th percentile and its BT utility are

$$b_n = \log(\ell_n + s_n^- \log 50), \quad u_n(r) = \sigma(r - b_n), \quad \sigma(t) = \frac{1}{1 + e^{-t}}. \quad (14)$$

Algorithm 2 Alignment: retained exponentiated-tail UCB–Pandora policy

Require: Frozen ρ from Eq. (9); divisor D ; at most 960 generations

- 1: Open the first three generations; set $n \leftarrow 3$
 - 2: **while** $n < 960$ **do**
 - 3: $E_j \leftarrow \exp(\text{clip}(R_j, -50, 50))$ for $j \leq n$
 - 4: Form T_n from the largest $n - \lceil 0.5n \rceil + 1$ values
 - 5: Compute (ℓ_n, s_n^+, s_n^-) by Eq. (13) with $(\kappa_r, \alpha, \delta) = (5, 0.4, 0.05)$
 - 6: Compute b_n , the 48-point $\widehat{\text{EI}}_n$, G_n , and $\widehat{\mu}_n^C$ by Eqs. (14)–(16)
 - 7: **if** $G_n \leq \widehat{\mu}_n^C/D$ **then**
 - 8: **break**
 - 9: **end if**
 - 10: Open generation $n + 1$ and observe (R_{n+1}, C_{n+1}) ; $n \leftarrow n + 1$
 - 11: **end while**
 - 12: **return** $(n, \max_{j \leq n} R_j, \sum_{j \leq n} C_j)$; this induces selection of an opened raw-reward maximizer and charges the exact character sum over all n opened responses
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Thus $u_n(r)$ is the win probability implied by a unit-temperature BT-style link against the estimated high-quality benchmark, not an empirically fitted probability calibration.

The implementation approximates the tail expected improvement with 48 fixed midpoint exponential quantiles. For $z_k = -\log(1 - (k + 1/2)/48)$, $k = 0, \dots, 47$, and raw incumbent $r_n^* = \max_{j \leq n} R_j$,

$$\widehat{\text{EI}}_n = \frac{0.5}{48} \sum_{k=0}^{47} [\sigma(\log(\ell_n + s_n^+ z_k) - b_n) - \sigma(r_n^* - b_n)]_+, \quad (15)$$

$$G_n = \min \left\{ \widehat{\text{EI}}_n + \eta \sqrt{\frac{\log(1/\delta)}{2n}}, 1 \right\}, \quad \widehat{\mu}_n^C = \frac{1}{n} \sum_{j=1}^n C_j. \quad (16)$$

The cost prior has zero weight, so $\widehat{\mu}_n^C$ is exactly the running mean of opened character counts. Algorithm 2 is the exact retained stopping calculation. Alignment uses one code-level divisor D . In the utility experiment it plays both roles, $D_{\text{res}} = D_{\text{util}} = D$; in head-to-head and target experiments it is only a reservation parameter.

How alignment is evaluated. For evaluation only, prompt i has benchmark $b_i^* = R_{i,(951)}$, the 951st smallest of its 960 rewards (the code’s fixed empirical 0.99-quantile convention), and stopped quality

$$Q_i(r) = \sigma(r - b_i^*). \quad (17)$$

This full-pool quantity never enters Algorithm 2; the algorithm uses the prefix benchmark b_n from Eq. (14).

Coding retains the same reservation comparison but changes the value map, tail support, and training priors.

1.5 Coding instantiation: shifted exponential in calibrated probability space

We condition on the 83 problems solved by at least one of 512 generations, then split this cohort into 41 training and 42 test problems. Conclusions therefore do not cover the excluded all-incorrect problems. Although the retained scripts also evaluate bounded-Gaussian and across-family series, all coding equations and reported results below isolate shifted-exponential policies in calibrated-probability space. Utility, equal-budget, and target quality use different pre-test selectors, which we describe separately rather than presenting them as one common stopped-policy series.

Train-only value calibration. For outer split s , one nondecreasing isotonic regression is fit directly to all 41×512 raw-reward/correctness pairs:

$$\widehat{g}_s \in \arg \min_{\substack{g: \mathbb{R} \rightarrow [0,1] \\ g \text{ nondecreasing}}} \sum_{i \in \mathcal{A}_s} \sum_{j=1}^{512} \{Y_{ij} - g(R_{ij})\}^2, \quad P_{ij} = \widehat{g}_s(R_{ij}) \in [0, 1]. \quad (18)$$

Here $Y_{ij} \in \{0, 1\}$ denotes correctness. Samples have equal weight; there is no reward binning. Predictions outside the training score range are clipped to the endpoint probabilities. The frozen map is applied to both training and test rewards, and Pandora utility is simply $u(R) = \widehat{g}_s(R)$. There is no BT transform in the retained coding results.

Distribution and character priors. For each prior-fitting problem i and tail quantile q , define

$$\ell_{i,q} = Q_q(P_{i1}, \dots, P_{i,512}), \quad (19)$$

$$s_{i,q} = \max\{\text{mean}_{j:P_{ij} \geq \ell_{i,q}}(P_{ij} - \ell_{i,q}), \epsilon\}. \quad (20)$$

The prior parameters are equal-problem averages

$$\bar{\ell}_q = \text{mean}_i \ell_{i,q}, \quad \bar{s}_q = \text{mean}_i s_{i,q}, \quad \bar{C} = \text{mean}_{i,j} C_{ij}. \quad (21)$$

During four-fold policy selection, Eq. (21) is refit on the three fit folds; for final evaluation it is refit on all 41 outer-training problems. The isotonic map in Eq. (18), however, is fit once on all 41 outer-training problems before those inner folds. Thus current-split test labels do not enter the isotonic fit, prior fit, or cross-fitted trajectories. The profiled utility and target selectors additionally use a frozen development profile built from earlier resamplings of the same 83-problem corpus; its provenance and resulting limitation are made explicit with those experiments below.

Online tail fit. For an opened prefix $P_1, \dots, P_n$, let

$$V_n = \max_{j \leq n} P_j, \quad \ell_n = Q_q(P_1, \dots, P_n), \quad (22)$$

$$I_n = \{j \leq n : P_j \geq \ell_n\}, \quad m_n = |I_n|, \quad (23)$$

$$s_n = \max\left\{\frac{1}{m_n} \sum_{j \in I_n} (P_j - \ell_n), \epsilon\right\}. \quad (24)$$

Location and scale are shrunk asymmetrically toward the training prior:

$$\tilde{\ell}_n = \frac{n\ell_n + \kappa_r \bar{\ell}_q}{n + \kappa_r}, \quad \tilde{s}_n = \frac{m_n s_n + \kappa_r \bar{s}_q}{m_n + \kappa_r}, \quad (25)$$

$$h_n = \sqrt{\frac{\log(1/\delta)}{m_n}}, \quad \ell_n^+ = \text{clip}_{[0,1]}(\tilde{\ell}_n + \alpha \tilde{s}_n h_n), \quad (26)$$

$$s_n^+ = \tilde{s}_n(1 + \alpha h_n). \quad (27)$$

The future upper tail has fixed mass $1 - q$. Conditional on being in that tail, X_n follows a shifted exponential with location ℓ_n^+ and scale s_n^+ , truncated and renormalized on $[\ell_n^+, 1]$:

$$f_n(x) = \frac{\exp(-(x - \ell_n^+)/s_n^+)}{s_n^+ \{1 - \exp[-(1 - \ell_n^+)/s_n^+]\}}, \quad \ell_n^+ \leq x \leq 1. \quad (28)$$

Consequently,

$$\widehat{\text{EI}}_n = (1 - q) \int_{\ell_n^+}^1 (x - V_n)_+ f_n(x) dx, \quad (29)$$

$$\hat{\mu}_n^C = \frac{\sum_{j=1}^n C_j + \kappa_c \bar{C}}{n + \kappa_c}, \quad G_n = \left(\frac{n}{3}\right)^{-\gamma} \widehat{\text{EI}}_n. \quad (30)$$

The code evaluates the integral using 32-point Gauss–Legendre quadrature through the inverse CDF of Eq. (28). Specifically, if $(x_k, w_k)_{k=1}^{32}$ are the Gauss–Legendre nodes and normalized weights on $[0, 1]$, and $M_n = 1 - \exp[-(1 - \ell_n^+)/s_n^+]$, the computed value is

$$\widehat{\text{EI}}_n = (1 - q) \sum_{k=1}^{32} w_k [\ell_n^+ - s_n^+ \log(1 - M_n x_k) - V_n]_+. \quad (31)$$

At $\ell_n^+ = 1$ the conditional law is interpreted as a point mass at one. Its tail mass remains exactly $1 - q$ even when isotonic plateaus make m_n larger than $(1 - q)n$.

Here α controls the optimism width. Positive α shifts the fitted tail upward, whereas $\alpha = 0$ leaves the prior-regularized plug-in fit. Both settings occur among selected policies. Tail decay and horizons below 512 are empirical regularizers outside classical Pandora stopping. Isotonic plateaus can tie several candidates; the implementation returns the first one attaining the maximum calibrated probability.

Having fixed the online rules, we next separate deployable policy evaluation from post-hoc budget- and quality-matched descriptions.

Algorithm 3 Coding: shifted-exponential Pandora in correctness-probability space

Require: Frozen isotonic map $\hat{g}_s$; prior $(\bar{\ell}_q, \bar{s}_q, \bar{C})$; parameters $(q, \kappa_r, \kappa_c, \alpha, \gamma, \delta, D_{\text{res}}, H)$

- 1: Open three generations; observe (R_j, C_j) and set $P_j \leftarrow \hat{g}_s(R_j)$ for $j = 1, 2, 3$; $n \leftarrow 3$
 - 2: **while** $n < H$ **do**
 - 3: Compute V_n , the local tail statistics, and (ℓ_n^+, s_n^+) by Eq. (27)
 - 4: Compute $\widehat{\text{EL}}_n$, $\hat{\mu}_n^C$, and G_n by Eqs. (28)–(30)
 - 5: **if** $G_n \leq \hat{\mu}_n^C / D_{\text{res}}$ **then**
 - 6: **break**
 - 7: **end if**
 - 8: Open generation $n + 1$; observe (R_{n+1}, C_{n+1})
 - 9: $P_{n+1} \leftarrow \hat{g}_s(R_{n+1})$; $n \leftarrow n + 1$
 - 10: **end while**
 - 11: **return** the first opened generation attaining $\max_{j \leq n} P_j$ and charge the exact prefix character sum
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1.6 What each experiment measures

The utility experiment evaluates Eq. (2) at a fixed economic price. Target-quality mode asks a different, constrained question: within a candidate policy class Π , use pre-test evidence to approximate

$$\hat{\pi}_\tau \in \arg \min_{\pi \in \Pi} \hat{C}_{\text{sel}}(\pi) \quad \text{subject to} \quad \hat{Q}_{\text{sel}}(\pi) \geq \tau. \quad (32)$$

Here quality and cost are aggregate means over tasks and randomized response orders. Consequently, τ is neither a per-task target nor a high-probability guarantee. Alignment quality is the relative BT-style score in Eq. (17); coding quality is observed binary correctness. Their target values are therefore not numerically comparable. In both tasks the selected divisor is a reservation parameter D_{res} , not an economic utility price.

Each target experiment reports transfer to held-out splits and two Fixed- N references. A training-selected Fixed- N rule targets τ using only pre-test data and is then frozen. A second Fixed- N reference is constructed after seeing the adaptive policy’s held-out quality; it controls for achieved quality when describing character efficiency, but it is an oracle diagnostic, not a deployable policy and not evidence that the requested target transferred.

1.6.1 Alignment experiments

The five alignment generators are Gemma-2-9B, Llama-3.1-8B, Llama-3.2-3B, Mistral-7B, and Qwen-2.5-7B. Each split uses four randomized training permutations and eight randomized test permutations per prompt. Every generator is scored by `mistral_rm_reward`, and a response’s character cost is Python’s `len` of its generated text (not token count, byte count, or prompt length). The three experiments reuse the same stopped prefixes but answer different questions:

1. **Utility objective.** At each reported divisor $D \in \{10^5, 5 \times 10^5, 10^6, 5 \times 10^6, 10^7, 5 \times 10^7\}$, training data choose the integer N maximizing $\text{mean}(Q) - \text{mean}(C)/D$. That N is frozen on test, and its utility is compared with Algorithm 2. If several values tie, the implementation takes the smallest maximizer.
2. **Equal-character head-to-head.** Each adaptive test trajectory is paired with an independently drawn opponent permutation. A randomized mixture of two adjacent Fixed- N policies is chosen so that its aggregate empirical mean character count exactly matches the adaptive mean; this is a mean-budget match, not a separate match for each trajectory. The reported score is $\Pr(R_{\text{adaptive}} > R_{\text{fixed}}) + \frac{1}{2} \Pr(\text{tie})$. Because the matching uses held-out outcomes, this is a descriptive equal-budget comparison, not a deployable selection rule.
3. **Target quality.** For requested aggregate BT-style evaluation qualities $\tau \in \{0.30, 0.35, \dots, 0.60\}$, training data bracket τ on the cumulative-maximum envelope of an 11-point adaptive reservation-divisor frontier. They form a convex mixture of adjacent policies; the same operation on the empirical training Fixed- N curve over $N = 1, \dots, 960$ forms a randomized adjacent- N comparator. Both mixtures are frozen before test. The primary cost-saving curve separately interpolates the held-out Fixed- N curve to exactly match the adaptive policy’s achieved aggregate held-out quality. This second comparison is a descriptive oracle. Twenty-eight of 350 generator–split–target rows lie below the cheapest adaptive frontier point and therefore target to $D_{\text{res}} = 50,000$: 27 at $\tau = 0.30$ and one at $\tau = 0.35$. The training Fixed- N curve brackets every target.

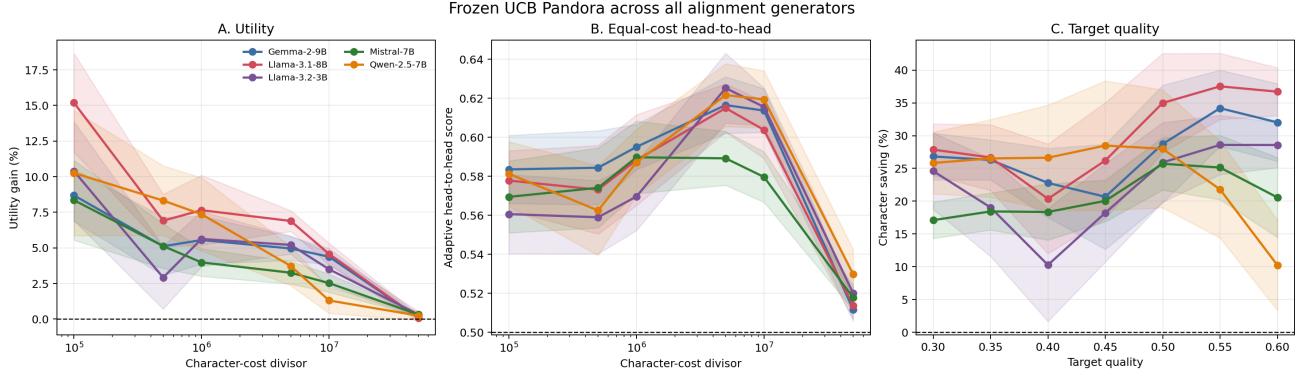


Figure 1: Alignment results over ten prompt splits for five generators. **A**: relative utility gain over the train-tuned, frozen Fixed- N policy. **B**: raw-reward head-to-head score against an independent adjacent- N mixture with exactly matched expected held-out characters; 0.5 is parity. **C**: character reduction relative to a post-hoc held-out Fixed- N interpolation matched to achieved quality. Bands are pointwise 95% Student- t intervals over ten split-level estimates. Panels B and C are controlled descriptive comparisons, not policy-selection procedures.

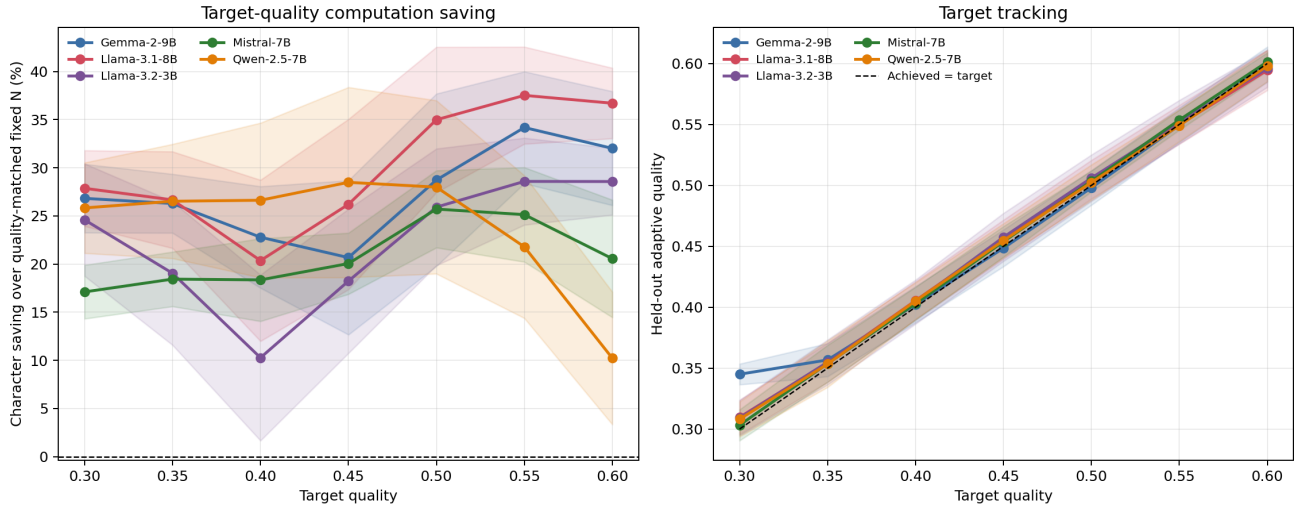


Figure 2: Alignment target-quality detail. The left panel repeats the post-hoc quality-matched oracle saving; the right panel shows transfer from requested aggregate BT-style quality to achieved held-out quality. The target is an aggregate mean, and the bands are pointwise Student- t intervals over overlapping split estimates.

Figure 1 shows the three questions side by side. The plotted statistics are first computed within each split and then averaged over the ten split estimates. Across $D \leq 10^7$, mean relative utility gains range from 1.31% to 15.19%, and all 25 generator-divisor pointwise intervals are above zero. At $D = 5 \times 10^7$ the gains are only 0.08–0.33%, and three of five intervals include zero. Every equal-character head-to-head interval is above 0.5. Quality-matched target savings range from 10.23% to 37.52%, with all 35 generator-target intervals above zero; these savings retain the oracle-matching caveat. Target transfer is not automatic: held-out quality reaches the requested threshold in 203 of 350 generator-split-target rows, with hit counts 36, 27, 28, 28, 30, 30, and 24 out of 50 as τ runs from 0.30 to 0.60. Low-end clamping produces the clearest overshoot: Gemma-2-9B achieves mean quality 0.3450 at requested quality 0.30. Against the frozen train-selected Fixed- N comparator, mean character savings range from -24.87% to 38.90% ; 30 intervals are above zero, four include zero, and the Gemma-2-9B/0.30 interval is below zero. This does not contradict the positive quality-matched curve: the direct comparison does not control for the adaptive policy’s held-out overachievement. The repeated 50/50 prompt splits overlap, so the Student- t bands summarize variation among split-level estimates; they are neither simultaneous bands nor intervals from ten independent datasets.

1.6.2 Coding calibration audit

Every coding protocol fits its isotonic map on the current split’s 41 outer-training problems. The saved calibration audits evaluate that frozen map on the 42-problem test half; those audit statistics do not enter the fitting call. Averaged over utility/equal-budget splits 75–84, outer-test correctness prevalence is 0.1344, mean predicted probability is 0.1354, and Brier score is 0.1050. On target splits 95–104, the corresponding values are 0.1346, 0.1270, and 0.1017. Similar marginal means are reassuring, but they are weaker evidence than a full reliability analysis; these audit records do not imply calibration at every probability level or a coverage guarantee.

1.6.3 Coding utility gap: the frozen profiled frontier

The authoritative coding utility result is the completed profiled-frontier experiment. For each economic divisor $D_{\text{util}} \in \{10^5, 2 \times 10^5, \dots, 10^6\}$, it chooses one policy configuration and one potentially different reservation divisor D_{res} . Its eligible policies are uncapped shifted-exponential rules with

$$q \in \{0.5, 0.75\}, \quad \alpha \in \{0.2, 0.8\}, \quad \kappa_r \in \{5, 20, 10^6\}, \quad \kappa_c \in \{0, 10\}, \quad \gamma \in \{0, 0.25, 0.5, 1\}, \quad H = 512. \quad (33)$$

There are 19 candidate reservation divisors. For every D_{util} , the frozen profile from split IDs 55–59 ranks candidates by $A_{\text{prof}} - C_{\text{prof,geo}}/D_{\text{util}}$ and supplies the best 20 to the next stage. Split IDs 65–69 evaluate these 200 candidates using 16 held-out permutations per problem. The selection score is

$$\frac{1}{2}U_{\text{prof}} + \frac{1}{2}\bar{U}_{\text{val}} - 0.1 \text{ sd}_{\text{val}}(U), \quad (34)$$

where the weights and penalty are chosen by leave-one-validation-split-out checks over the recorded meta-grid. One row per D_{util} is then frozen.

On final split IDs 75–84, each frozen rule is evaluated on 42 test problems with 48 permutations per problem. The current split’s 41 training problems fit only its isotonic map, tail/character prior, and Fixed- N comparator; they do not retune the frozen configuration or D_{res} . The online rule is exactly

$$\left(\frac{n}{3}\right)^{-\gamma} \widehat{\text{EI}}_n > \frac{\hat{\mu}_n^C}{D_{\text{res}}}, \quad U_{\text{test}} = \text{correctness} - \frac{\text{exact opened characters}}{D_{\text{util}}}. \quad (35)$$

All ten frozen choices have $\alpha > 0$, $\gamma > 0$, and no Fixed- N cap; nine use $\gamma = 1$ and one uses $\gamma = 0.5$.

Table 2: Final coding utility-gap results for the frozen profiled frontier. Additive-gap intervals are pointwise 95% Student- t intervals over ten split estimates.

D_{util}	D_{res}	Relative gain (%)	Additive gap [95% CI]	Positive splits
100k	70k	9.49	.01927 [.01209, .02644]	10/10
200k	70k	4.90	.01132 [.00773, .01491]	10/10
300k	400k	2.50	.00598 [.00034, .01161]	9/10
400k	400k	1.37	.00289 [−.00275, .00852]	8/10
500k	400k	3.42	.00843 [.00149, .01537]	9/10
600k	400k	2.84	.00662 [−.00153, .01476]	8/10
700k	1m	1.34	.00240 [−.00773, .01253]	7/10
800k	1m	1.14	.00188 [−.00778, .01155]	7/10
900k	1m	1.15	.00195 [−.00823, .01213]	6/10
1m	1m	0.93	.00143 [−.00906, .01192]	6/10

The mean relative gain is positive at every reported price, and 80 of 100 split–divisor gaps are positive. The simple mean of the 100 splitwise relative gains is 2.91%. Uncertainty remains material: the additive-gap interval excludes zero only at 100k, 200k, 300k, and 500k. Relative to the older retained shifted-exponential selector, the new mean is higher at all ten divisors and removes four negative mean points, but the experiment does not report paired intervals for that between-policy improvement.

The development, validation, and final stages use distinct split IDs and permutations, but not disjoint problem identities. Development held-out halves cover 81 of 83 problems, validation held-out halves cover 80, and their union covers all 83; every problem in every final test half appears in that earlier union. Hyperparameters are frozen before the final split aggregates and permutations are evaluated, yet this remains within-corpus repeated-split evidence rather than a test on new problem identities. The pointwise intervals likewise summarize overlapping resamples, not independent datasets or external-population uncertainty.

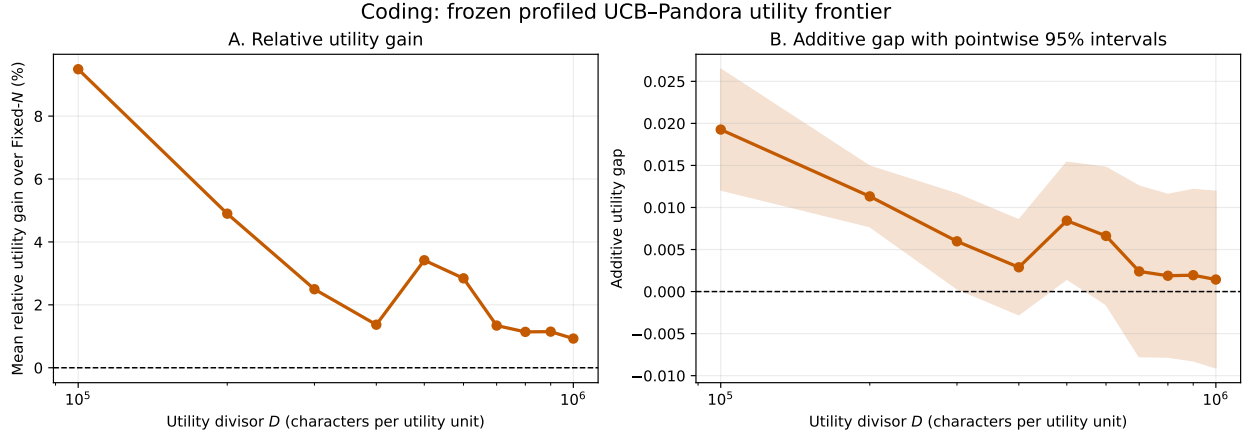


Figure 3: Coding utility results for the globally frozen profiled frontier. The left panel shows mean splitwise relative gain over the current split’s train-tuned Fixed- N comparator; the right panel shows the additive utility gap and pointwise Student- t intervals. Every trajectory is charged its exact opened characters at D_{util} .

1.6.4 Coding equal-budget diagnostic: the retained selector

The improved frontier does not produce an equal-budget opponent. That diagnostic therefore comes from the earlier shifted-exponential experiment on the same final split IDs 75–84. Separately for each outer split and divisor, four-fold training selection searches

$$\begin{aligned} q \in \{0.5, 0.75\}, \quad \alpha \in \{0, 0.2, 0.8\}, \quad \kappa_r \in \{5, 20, 10^6\}, \\ \kappa_c \in \{0, 10\}, \quad a \in \{1, 1.1, 1.2, 1.3, 1.5, 2, \infty\}, \quad \gamma = 0, \end{aligned} \quad (36)$$

using four permutations per training problem. A finite a sets $H = \min\{512, \max(3, \lceil aN_D^* \rceil)\}$, where N_D^* is the smallest maximizer of training Fixed- N utility. The isotonic map and N_D^* use all 41 outer-training problems; only the distribution and character priors are refit inside the four folds. Of the 100 saved choices, 91 use a finite cap, 56 have $\alpha > 0$, and 44 use $\alpha = 0$. Within each outer split and divisor, the selector chooses the configuration that maximizes four-fold cross-fitted training utility: observed accuracy minus opened characters divided by the shared utility/reservation divisor $D_{\text{util}} = D_{\text{res}} = D$.

On 48 final-test permutations per problem, an independent opponent permutation and a post-hoc adjacent- N mixture exactly match the adaptive policy’s aggregate mean characters. This comparison isolates accuracy at a common measured budget but is descriptive because held-out costs determine the mixture. The shifted-exponential accuracy advantage is 1.51 percentage points [0.55, 2.47] at 100k, 1.19 [0.41, 1.98] at 200k, 0.73 [0.17, 1.28] at 300k, and 0.68 [0.24, 1.11] at 400k. The remaining six pointwise intervals include zero. These numbers belong to this retained selector and must not be attributed to the improved utility-frontier policies.

1.6.5 Coding target-quality experiment

For final split IDs 95–104, four-fold selection uses 16 training permutations per problem and final evaluation uses 48 test permutations per problem. Here $\tau \in \{0.25, 0.30, 0.35\}$ targets aggregate mean *observed binary correctness*, not mean isotonic predicted probability. Isotonic calibration supplies the online value scale; it is not the target-attainment criterion. Separately on each outer split, the selector chooses one uncapped shifted-exponential configuration and one of 19 reservation divisors from

$$q \in \{0.5, 0.75\}, \quad \alpha \in \{0, 0.2, 0.8\}, \quad \kappa_r \in \{5, 20, 10^6\}, \quad \kappa_c \in \{0, 10\}, \quad \gamma \in \{0, 0.25, 0.5, 1\}, \quad H = 512. \quad (37)$$

For policy h and reservation divisor D_{res} , a frozen profile from split IDs 55–59 stores the arithmetic mean of held-out observed binary accuracy $A_{\text{prof}}(h, D_{\text{res}})$ and geometric-mean characters

$$C_{\text{prof}}(h, D_{\text{res}}) = \exp \left\{ \frac{1}{5} \sum_{s=55}^{59} \log C_s(h, D_{\text{res}}) \right\}. \quad (38)$$

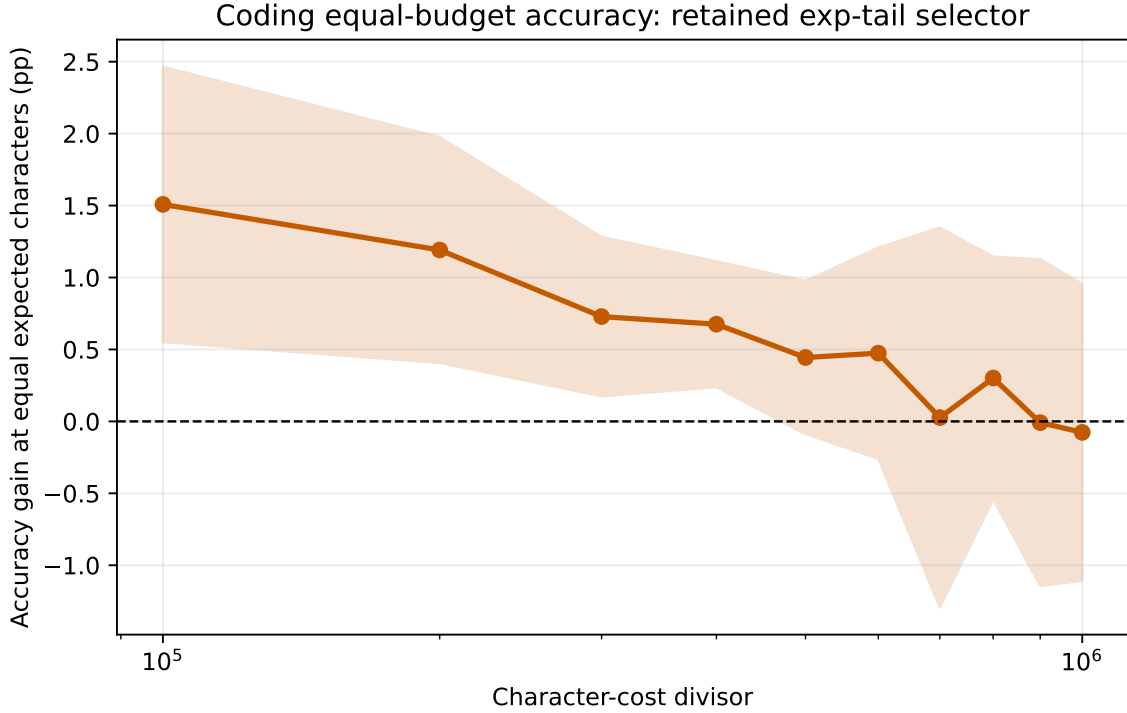


Figure 4: Equal-budget coding accuracy for the retained per-split shifted-exponential selector. The adjacent- N opponent is fitted post hoc to held-out mean characters; bands are pointwise Student- t intervals over overlapping split estimates.

Here $A_{CF,s}$ is the four-fold cross-fitted observed accuracy on the current outer-training split. The feasibility score is

$$A_{sel,s}(h, D_{res}) = 0.1A_{CF,s}(h, D_{res}) + 0.9A_{prof}(h, D_{res}). \quad (39)$$

Among candidates with $A_{sel,s} \geq \tau$, selection minimizes frozen C_{prof} , with smaller overshoot and then fixed identifiers breaking ties. Current-split character cost receives zero selection weight. The implementation has a cheapest-max-score fallback if no candidate is feasible, but it was inactive because all 30 retained choices passed their selection threshold. There is no policy mixture, target margin, lower-confidence threshold, Fixed- N guard, or test-time selection. All 30 selected policies have $\gamma > 0$; 27 have $\alpha > 0$ and three have $\alpha = 0$.

The profile is pre-test with respect to each current final split’s aggregate outcomes and response permutations, but it is not identity-disjoint. Its held-out halves contain 81 of the 83 problems, and 40–41 members of each 42-problem final-test half occur in at least one earlier development half. Target results are therefore within-corpus repeated-split evidence, not generalization to unseen problem identities. The exact profile CSV and hash are supplied, so selection can be replayed; the five diagnostic NPZ files from which that profile was built are absent, so the profile itself cannot be reconstructed from the snapshot alone.

Two integer Fixed- N baselines answer different questions. The deployable baseline is the cheapest training N whose exact training accuracy reaches τ , frozen before test. If no N reaches the target, the code uses the cheapest N on the maximum-accuracy plateau. The descriptive oracle is the cheapest held-out integer N whose exact expected accuracy is at least the *adaptive policy’s achieved held-out accuracy*; because N is integer, it generally overshoots rather than exactly matches. The same maximum-plateau fallback applies when the threshold is unattainable. At $\tau = 0.35$, one of ten shifted-exponential splits invokes this fallback for both baselines. Adaptive results average 48 sampled permutations; both Fixed- N accuracy curves are exact without-replacement expectations with the implementation’s tied-score handling.

Panel B reports aggregate ratios, not arithmetic means of per-split percentages. For adaptive characters $C_{A,s}$ and baseline characters $C_{B,s}$ —using either the train-selected Fixed- N or held-out integer- N reference—the reported saving is

$$100 \left(1 - \frac{\text{mean}_s C_{A,s}}{\text{mean}_s C_{B,s}} \right), \quad (40)$$

with a paired 50,000-resample bootstrap over splits for each baseline.

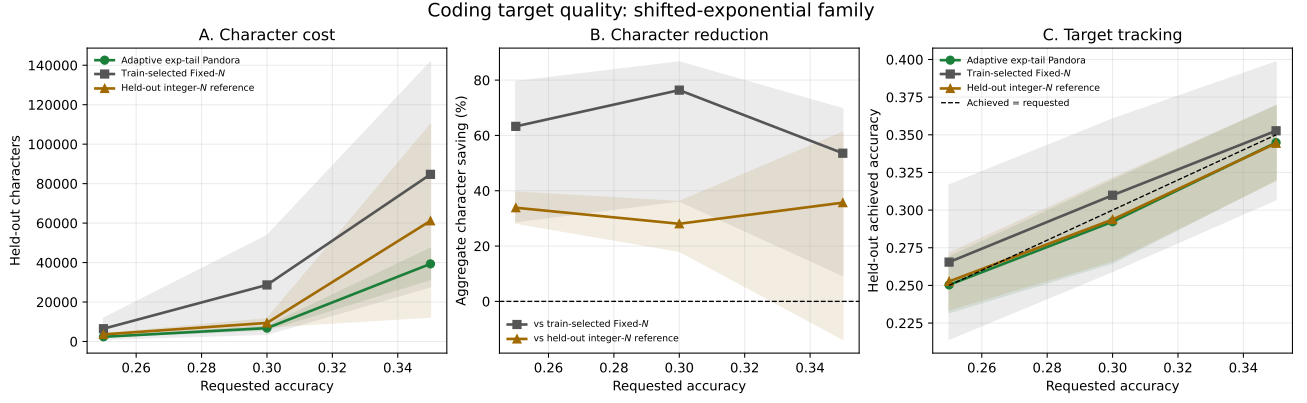


Figure 5: Coding target-quality results for the shifted-exponential family. **A:** held-out character cost for the adaptive policy and the two integer Fixed- N references. **B:** aggregate character savings; the held-out reference is the first integer N reaching at least the achieved adaptive accuracy when attainable, with one $\tau = 0.35$ fallback. **C:** achieved versus requested accuracy. The frozen development profile reuses the same finite problem corpus, as detailed in the text.

Table 3: Coding target-quality results for the shifted-exponential family. Accuracy intervals are pointwise Student- t intervals; saving intervals use the paired split bootstrap.

τ	Achieved accuracy [95% CI]	Saving vs train Fixed- N	Saving vs held-out integer- N	Target hits
.25	.2503 [.2319, .2688]	63.28% [28.84, 79.46]	33.86% [28.29, 39.52]	5/10
.30	.2923 [.2648, .3197]	76.38% [36.30, 86.56]	28.06% [18.12, 36.09]	3/10
.35	.3448 [.3203, .3694]	53.57% [9.35, 69.65]	35.69% [-13.69, 61.15]	3/10

The train-selected Fixed- N rules achieve held-out accuracies 0.2655, 0.3099, and 0.3527 and hit the requested threshold in 6/10, 6/10, and 4/10 splits, so their larger savings are not quality-matched comparisons. The held-out integer- N threshold is attained in 10/10, 10/10, and 9/10 splits. The adaptive selector tracks the target fairly closely in the aggregate but offers no per-split attainment guarantee. The bootstrap likewise resamples ten overlapping partitions of the same corpus; its interval is a descriptive repeated-split uncertainty summary, not an external-population interval.

1.7 Relation to unknown-distribution, unknown-cost Pandora’s box

Viewed through the reservation-value lens, the two systems have the same structure. Before opening, both the next score and the next response length are unknown. After a three-response pilot, each method fits a training-regularized upper tail to the observed score proxy, estimates the next length, and opens another response only while adjusted one-step improvement exceeds estimated character cost. Alignment realizes this blueprint with an optimistic shifted-exponential tail in exponentiated reward space and a moving BT-style benchmark. Coding uses a shifted-exponential tail in frozen isotonic probability space, with optional character-prior shrinkage and tail decay. The primary coding utility frontier additionally calibrates its reservation divisor separately from its reporting price and is uncapped; the retained equal-budget diagnostic is the protocol that sometimes uses a training-derived hard horizon.

We therefore use *unknown-distribution, unknown-next-cost UCB-Pandora family* as the umbrella description. Here “UCB” means a UCB-shaped optimistic index, not a proved finite-sample confidence bound or regret guarantee. The alignment rule always uses a positive deformation. Coding’s $\alpha > 0$ configurations do too, whereas $\alpha = 0$ is the zero-width, prior-regularized plug-in boundary of the same implementation. Every frozen policy in the primary coding utility frontier has $\alpha > 0$; three of 30 coding target selections and 44 of 100 policies in the separate equal-budget protocol have $\alpha = 0$.

Finally, these are empirical reservation policies, not claims of Weitzman-optimal behavior. The pools are finite and sampled without replacement; tails may be misspecified; coding correctness remains latent; reward-length dependence is ignored; and the regularizers, reservation calibration, and equal-budget caps fall outside classical assumptions. These caveats bound the theoretical claim but do not change the exact implementation mapping to Algorithm 1.